

AI2002 – Artificial Intelligence (BDS-5A)

Project

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Instructions:

1. Only One member (Lead) for each group will submit.
2. Project ID will be assigned to each Project, your submission file Name should be your project. Each of which starts with F26-id. The purpose of this is to get you familiar with FYP submission.
3. It should be clear that your Project will not get any credit if:
 - The Project is submitted after the due date.
 - Project is plagiarized from any source. (Online/other students/etc.)

Students must be able to explain and justify all submitted work during evaluation.

General Guidelines

- Group size: 3 students
 - Each group must register their project details in the provided sheet
 - Each group must select one project track only
 - Each group must propose a unique problem / paper
 - Any external code, dataset, or model must be properly cited
 - Respect due date of every deliverable
 - Marks will be awarded mainly for the AI-related backend logic and analysis. User interface design is secondary and can only contribute toward bonus marks.
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1 Objective

The primary objective of this course project is to provide students with a comprehensive, hands-on experience in the lifecycle of an Artificial Intelligence project. Beyond theoretical understanding, students will navigate the practical challenges of finding real-world datasets, preprocessing raw information, training robust models, and either building functional user interfaces or conducting rigorous scientific analysis.

2 Track Selection

Students must choose one of the following two tracks:

2.1 Track A: Research & Development (R&D)

Goal: Investigate a specific AI hypothesis or optimize an architecture for a niche task.

- **Focus:** Comparative analysis, ablation studies, and technical novelty.
- **Deliverable:** A rigorous technical paper and a codebase for reproducibility.

2.2 Track B: Research Paper Implementation

Goal: Reproduce and analyze a published research paper.

- **Focus:** Code execution, reproducibility, and comparative benchmarking.
 - **Deliverable:** Successful execution of paper code and a report comparing results with baseline models.
 - **Resources:** Students may look for relevant papers on Hugging Face. Note: Using this specific site is not mandatory.
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3 Project Workflow

3.1 Phase 1: Problem Definition / Paper Selection

- **Track A:** Identify a domain and a specific problem or research gap.
- **Track B:** Select a relevant AI research paper. Students must justify the selection and explain the core contribution of the paper.

3.2 Phase 2: Data Acquisition & Setup

- **Track A:** Gather data via scraping, APIs, or repositories. Clean and preprocess as required.
- **Track B:** Obtain the dataset used in the paper. If the dataset is private, identify a suitable open-source proxy dataset for testing.

3.3 Phase 3: Implementation & Execution

- **Track A:** Choose appropriate algorithms based on the problem. Use standard frameworks such as PyTorch, TensorFlow, or Scikit-Learn.
- **Track B:** Execute the paper's code. If the architecture is too advanced for available hardware, partial execution with a deep understanding of the modules is also accepted.

3.4 Phase 4: Evaluation & Comparative Analysis

- **Track A:** Evaluate using relevant metrics (Accuracy, F1, RMSE, etc.) and perform a bias check.
- **Track B:** Compare the proposed paper model results with baseline models. Analyze and explain why the proposed model outperformed the baselines.

3.5 Phase 5: Output (Analysis)

- **Track A & B:** A technical report documenting experimental results, error analysis, and theoretical insights.

3.6 Phase 6: Documentation & Final Report

Prepare a technical report (in LaTeX) covering:

- Methodology and architecture diagrams.
 - Challenges faced during implementation (e.g., environment setup, training convergence).
 - **Track B:** A specific section on “Replication Results” vs. “Reported Paper Results.”
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4 Grading Rubric

Criteria	Track A	Track B
Originality / Paper Selection	25%	15%
Data Preprocessing / Setup	15%	20%
Technical Implementation	30%	35%
UI / Experimental Rigor	20%	20%
Documentation & Report	10%	10%

BONUS: A neat user-friendly interface will land you bonus points!!!

Good Luck !